

ENGEN370-23A Project Brief

Group 07

“With recent bans in using cheap immigrant labour for agriculture and forestry, the industries no longer have the human resources for planting, maintenance, thinning and harvesting, and are not willing (or cannot afford) to pay appropriate wages to attract residential labour. Robotics are being implemented to replace manual labour. What is the environmental and social impact of switching to robotics versus maintaining manual labour and potential issues around resilience?”

1. Societal Demand

Our societal demand is the universal and never-ending need for nutrition provided by foodstuffs from the agricultural industry. This project will be addressing the problem of labour shortages in the New Zealand Kiwifruit industry in regards to seasonal manual labour. Our solutions will seek to use robotics to assist or replace the need for manual labour in this industry.

We are focusing on the societal demand of nutrition because it is one that will never go away and is created by all people. Simply put, people need food to survive. More than that, they need efficiently produced food, to support a growing population in an environment of low labour supply. Kiwifruit specifically is not considered a staple food but participates in a healthy balanced diet of fruit, vegetables, meat and carbohydrates.

2. Solutions to the Demand

2.1 Manual Labour (Abbas Abusaleh)

This solution uses ‘manual labour’ to pick kiwifruit. This solution involves people such as retirees, seasonal workers, and students being recruited to harvest the kiwifruit grown (Garcia Contracting Services, n.d.). While expanding the pool of available labourers may prove challenging, this solution increases the productivity of current employees through assisting tools like a chest-mounted bag to make labour easier. Rather than purchasing and maintaining expensive equipment, it is most common to use this solution (Garcia Contracting Services, n.d.). By meeting the public demand for kiwifruit, this solution can meet the societal demand for nutrition.

2.2 Unsupervised Automatic Picker (Alexander Stokes)

Robotics Plus, a New Zealand company, is developing a completely autonomous kiwifruit picker. The system uses computer vision and four robotic arms to pick, store, and offload its kiwifruit without direct supervision (Scarfe, 2012).

This product is not yet commercially available, but it has been noted to be more consistent than people and provide better quality fruit (Robotics Plus, 2016). It seeks to replace, or supplement, human manual labour for picking kiwifruit and therefore reducing wasted produce (Robotics Plus, 2021). By meeting the demand for kiwifruit, this solution can meet the societal demand for nutrition.

2.3 Supervised Automatic Picker (Liam Labuschagne)

FFRobotics is developing an apple picker, which involves hydraulic-powered gripper hands. The selling point is the ability for multiple hands to operate simultaneously. While not specifically for Kiwifruit, we can adapt it to work for kiwifruit.

FFRobotics estimates that its system can pick up to 10,000 apples per hour. This system picks fruit autonomously, using AI to select fruit that is optimal for the market and sorts it. This solution can meet the societal demand for nutrition by applying this technology to kiwifruit, and picking fruit that would otherwise be left to waste due to a shortage of labour.

2.4 Assistive Picker (Caleb Archer)

University of Waikato researchers, in collaboration with Zespri and Garcia Contracting Services, have developed an e-Bin device to assist with the kiwifruit picking process and reduce shortage impacts. The e-Bin is intended to augment the abilities of human pickers (Hunter, Z., 2022).

The difficulty of the manual picking process further provokes the kiwifruit industry's labour shortages. Specifically, having to carry heavy loads (of up to 25 kilograms) of picked kiwifruit eliminates less physically able individuals as workers. The e-Bin acts as a moving storage unit for four workers at a time, removing the use of an individual pack.

3. Case Study

The desire and need for continuous food production is an ever-present issue within the kiwifruit industry. Garcia Contracting Services provides kiwifruit orchard services to orchards within the Bay of Plenty, the largest kiwifruit production region in New Zealand (Tupu.nz, n.d.). As the provider and supervisor of workers sent to these orchards, Garcia is impacted by labour shortages (Garcia Contracting Services, n.d.).

Even prior to the border closures and Recognised Seasonal Employer worker restrictions during the pandemic, the perceived working conditions of the kiwifruit industry have contributed to the difficulty of recruiting and retaining staff available to perform orchard maintenance and fruit production tasks (Hunter, Z., 2022). The manual nature of orchard work fails to motivate potential workers.

Solutions that have been discussed, and will be elaborated on, shall deal with the kiwifruit picking process and its potential for partial or full automation with the use of robotics.

4. Client

We have chosen New Zealand kiwifruit exporter Zespri as a client. We chose Zespri because it is New Zealand's sole global exporter to all countries (except Australia) (Zespri, n.d.). Zespri is very influential, as it partners with thousands of individual kiwifruit orchards, nationally and internationally, to source their kiwifruit.

5. References

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